

DW144/TR19 - DUCTWORK ACCESS HATCHES (SAME TABLE IN BOTH DOCUMENTS)

NOTES FROM DW144 RE. TABLE 20 -

WHILST IT SHALL BE STANDARD PRACTICE TO PROVIDE ACCESS PANELS FOR THE INSPECTION AND SERVICING OF PLANT AND EQUIPMENT AS DETAILED IN 20.2 ABOVE, ADDITIONAL PANELS MAY BE REQUIRED FOR CLEANING INSPECTION AND CLEANING ACCESS AND THESE WILL BE SITE-FITTED WHERE NECESSARY BY A SPECIALIST CLEANING CONTRACTOR. TABLE 20 BELOW IS A SUMMARY OF THE PARTIES RESPONSIBLE FOR FITTING ACCESS PANELS EITHER FOR INSPECTION/SERVICING ACCESS TO INLINE EQUIPMENT / COMPONENTS OR CLEANING INSPECTION AND CLEANING ACCESS. THIS TABLE IS REPRODUCED IN FULL IN BESA PUBLICATION TR/19 'GUIDE TO GOOD PRACTICE, INTERNAL CLEANLINESS OF VENTILATION SYSTEMS' WHICH COVERS THE SUBJECT OF CLEANING INSPECTION AND CLEANING ACCESS IN GREATER DETAIL.

TABLE 20 (DW144)/ TABLE 3 (TR19) - SAME INFORMATION IN BOTH DOCUMENTS LOCATION OF ACCESS PANELS FOR INSPECTION/SERVICING AND OR INTERNAL CLEANLINESS PURPOSES			
INLINE EQUIPMENT	LOCATION	DUCTWORK CONTRACTOR	SPECIALIST CLEANING CONTRACTOR
VOLUME CONTROL DAMPER	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
FIRE DAMPER	BOTH SIDES	TO SUIT DAMPER MAINTENANCE	OPPOSITE
HEATING/COOLING/RECLAIM COILS	BOTH SIDES	PANEL ON BOTH SIDE	
ATTENUATORS (RECTANGLES)	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
ATTENUATORS (CIRCULAR)	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
FILTER SECTION	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
AIR TURNING VANES	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
CHANGES OF DIRECTION	ONE SIDE		ONE PANEL TO SUIT
IN-DUCT FANS/DEVICES	BOTH SIDES	UP-STREAM PANEL	DOWN-STREAM PANEL
INLET/EXHAUST LOUVRE	ONE SIDE	ONE PANEL TO SUIT	
INTERMEDIATE CLEANING PANELS			TO SUIT FREQUENCY SPECIFIED IN TR19 & DW172

UNLESS SPECIFICALLY STATED IN THE TENDER DOCUMENTATION, KEY IS/ DUCTWORK INSTALLERS WILL PROVIDE THE ACCESS HATCH REQUIREMENT AS NOTED UNDER DUCTWORK CONTRACTOR - HIGHLIGHTED IN RED.

ADDITIONAL HATCHES REQUIRED BY SPECIALIST CLEANING CONTRACTOR PROVIDED BY OTHERS

NOTES TO TABLE 20 (DW144)

SYSTEM DESIGNERS MUST ENSURE THAT CONSIDERATION IS GIVEN TO THE FOLLOWING ASPECTS:

- PROVIDING SUFFICIENT CLEARANCE FOR CLEANING ACCESS BETWEEN IN-LINE ITEMS OF EQUIPMENT, E.G.BETWEEN ATTENUATORS AND FANS.
- IN ORDER TO FIT ACCESS PANELS, PROVIDING SUFFICIENT SPACE ON THE SURFACES OF LARGE FITTINGS SUCH AS, FOR EXAMPLE, SEGMENTED BENDS IN SPIRALLY WOUND SYSTEMS.
- THE IMPORTANCE OF THERE BEING SUFFICIENT CLEARANCE FOR CLEANING ACCESS BETWEEN AN INLET/EXHAUST LOUVRE AND AN ITEM OF INLINE PLANT EQUIPMENT SUCH AS AN ATTENUATOR OR FAN.
- RECOGNISING THAT FIRE RESISTING KITCHEN EXTRACT SYSTEMS NEED A GREATER FREQUENCY OF ACCESS PANELS AS DETAILED IN THE BESA PUBLICATION DW/172, 'SPECIFICATION FOR KITCHEN VENTILATION SYSTEMS' AND THAT SUCH PANELS MUST BE INCORPORATED INTO THE MANUFACTURE OF THE KITCHEN EXTRACT DUCTWORK. IN THE CASE OF FIRE-RESISTING DUCT SYSTEMS PARTICULAR CARE MUST BE TAKEN TO ENSURE THAT ANY RETRO-FITTED ACCESS PANELS ARE SUITABLY FITTED, UNDER LICENCE TO THE FIRE RESISTING SYSTEMS MANUFACTURER.

NOTES TO TABLE 3 (TR19)

- WITH REGARD TO NEW DUCTWORK INSTALLATIONS DW/144 (2013) TABLE 20 (*SAME AS TABLE 3 ABOVE*) SPECIFIES THE LOCATION OF AND RESPONSIBILITY FOR THE INSTALLATION OF ACCESS DOORS.
- OTHER THAN IN THE LOCATIONS SHOWN IN TABLE 3, INTERMEDIATE ACCESS PANELS SHOULD BE INSTALLED BY THE SPECIALIST CLEANING CONTRACTOR AS A MINIMUM EVERY 15 M IN HORIZONTAL DUCTWORK SYSTEMS FOR THE PURPOSE OF NORMAL USAGE BUT NOT GREATER THAN 1M FROM THE THROAT OF A SQUARE 90° BEND.
- IN THE CASE OF FLAT OVAL DUCTWORK INCORPORATING INTERNAL TIE RODS IT MAY BE NECESSARY TO INSTALL ACCESS DOORS AT MORE FREQUENT INTERVALS. IN THE CASE OF VERTICAL DUCTWORK, AN ACCESS OPENING AT THE BOTTOM AND TOP OF EACH RISER AND AT EACH ACCESSIBLE FLOOR SHOULD BE PROVIDED FOR CLEANING AND INSPECTION PURPOSES.
- ACCESS PANELS CAN BE POSITIONED TO ALLOW ACCESS TO MORE THAN ONE COMPONENT DEPENDING ON DUCTWORK DIMENSIONS, DISTANCE BETWEEN COMPONENTS AND THE CLEANING TECHNIQUE USED.
- ACCESS PANELS SHOULD NORMALLY BE POSITIONED AS CLOSE AS PRACTICABLY POSSIBLE TO THE ITEM TO BE CLEANED (0.5 M MAXIMUM).
- A CHANGE OF DIRECTION IS DEEMED TO BE A 90° BEND AND BRANCHES. WITH REGARD TO CIRCULAR DUCTWORK IT IS NOT NECESSARY FOR THERE TO BE AN ACCESS PANEL ON EVERY CHANGE OF DIRECTION.
- FOR ACCESS REQUIREMENTS FOR KITCHEN EXTRACT SYSTEMS SEE SECTION 7.
- RECOGNISING THAT FIRE RESISTING KITCHEN EXTRACT SYSTEMS NEED A GREATER FREQUENCY OF ACCESS PANELS AS DETAILED IN THE BESA PUBLICATION DW/172, 'SPECIFICATION FOR KITCHEN VENTILATION SYSTEMS' AND THAT SUCH PANELS MUST BE INCORPORATED INTO THE MANUFACTURE OF
- THE KITCHEN EXTRACT DUCTWORK. IN THE CASE OF FIRE-RESISTING DUCT SYSTEMS PARTICULAR CARE MUST BE TAKEN TO ENSURE THAT ANY RETRO-FITTED ACCESS PANELS ARE SUITABLE FITTED, UNDER LICENCE TO THE FIRE-RESISTING SYSTEMS MANUFACTURER.

TR19 GREASE - ACCESS DOORS IN KITCHEN DUCTWORK

TABLE 1 -LOCATION OF ACCESS PANELS FOR CLEANING AND INSPECTION PURPOSES	
TYPE	POSITION
VOLUME CONTROL DAMPER	BOTH SIDES
FIRE DAMPER	BOTH SIDES
ATTENUATORS	BOTH SIDES
CHANGES IN DIRECTION	BOTH SIDES
FILTER SECTIONS	BOTH SIDES
HORIZONTAL DUCTS	GENERALLY 2m CENTRES (SEE NOTE 2)
RISERS	MAXIMUM 2m CENTRES (SEE NOTE 3)
EXTRACT FANS	BOTH SIDES (SEE NOTE 4)
DISCHARGE GRILLE/MESH	ONE SIDE (SEE NOTE 5)

TR19 - ACCESS DOORS SIZES

TABLE 4 -RECOMMENDED SIZE OF OPENING (RECTANGULAR & FLAT OVAL)		
DUCT SIZE UP TO LONGEST SIDE MAJOR AXIS	RECOMMENDED DIMENSIONS OF OPENING	
200 mm	300 mm	150 mm
300 mm	300 mm	200 mm
400 mm	450 mm	300 mm
≥500 mm	450 mm	450 mm

TABLE 5 -RECOMMENDED SIZE OF OPENING (CIRCULAR)		
DUCT SIZE UP TO	RECOMMENDED DIMENSIONS OF OPENING	
310 mm	250 mm	150 mm
400 mm	400 mm	300 mm
500 mm	400 mm	300 mm
≥600 mm	500 mm	400 mm

NOTES TO TR-19 TABLE 1: (taken direct from TR-19, struck through where not applicable)

- ADDITIONAL BUILDERS WORK HATCHES MAY NEED TO BE FITTED IN THE CEILINGS AND/OR WALLS OF EXISTING INSTALLATIONS, OR PROVIDED FOR IN NEW CONSTRUCTIONS. THESE WOULD NOT BE PROVIDED BY THE CLEANING CONTRACTOR UNLESS SEPARATELY SPECIFIED AND PRICED.DUCTWORK.
- ACCESS OPENINGS FOR CLEANING PURPOSES ARE GENERALLY REQUIRED AT A MAXIMUM OF 2 METRE CENTRES AND/OR AT EACH CHANGE OF DIRECTION TO ENABLE HEAD AND SHOULDER ACCESS. THIS DISTANCE SHOULD BE REDUCED WHERE THE SIZE OF THE DUCT PREVENTS ADEQUATE CLEANING BY HAND, WHERE THERE ARE SEVERAL CHANGES IN DIRECTION OR WHERE OTHER EXTERNAL FEATURES RESTRICT THE POSITIONING OF PANELS.
- INTERNAL KITCHEN EXTRACT RISERS OFTEN REQUIRE ACCESS PANELS FITTED AT MAXIMUM 2 METRE CENTRES ON EACH FLOOR LEVEL SO THAT ALL INTERNAL SURFACES CAN BE REACHED AND FIRE DAMPERS (*SUPPLY DUCT ONLY*), WHERE FITTED, CAN BE CLEANED. IN SOME BUILDINGS THIS MAY REQUIRE ADDITIONAL BUILDERS WORKS (E.G. HATCHES THROUGH MASONRY/STONE/CONCRETE) TO REACH THE RISER DUCTS (SEE NOTE 1). THE EXCEPTION TO THIS MAY BE IF MECHANIZED REMOTE CLEANING OR ABSEIL CLEANING IS PRACTICABLE, WHERE SPECIALIST ADVICE SHOULD BE SOUGHT. IN SUCH CASES IT MAY BE POSSIBLE TO REDUCE THE REQUIRED QUANTITY OF ACCESS PANELS, SUBJECT TO ENSURING THAT GREASES LEVELS PRE AND POST CLEANING CAN BE ADEQUATELY MONITORED.
- EXTRACT FAN DESIGN SHOULD ALLOW THOROUGH CLEANING OF IMPELLER BLADES AND INTERNAL SURFACES WITHOUT THE NEED FOR DISMANTLING, I.E. DUCTWORK WITH ACCESS PANELS SHOULD BE PROVIDED IMMEDIATELY UPSTREAM AND DOWNSTREAM OF THE FAN. LARGER FANS SHOULD BE DESIGNED WITH PANELS IN THE CASING. SIMILARLY, ATTENUATORS OR OTHER IN-LINE FITTINGS LIKELY TO OBSTRUCT CLEANING ACTIVITY MUST BE PROVIDED WITH ADEQUATELY SIZED ACCESS PANELS ON BOTH SIDES.
- AT THE DESIGN STAGE, CONSIDERATION SHOULD BE GIVEN TO THE PROVISION OF SAFE ACCESS TO THE DOWNSTREAM SIDE OF DISCHARGE GRILLES, BIRD GUARD MESH AND LOUVRES. ATTENUATORS SHOULD ALSO BE GIVEN SPECIAL CONSIDERATION FOR ACCESS.
- AT THE DESIGN AND INSTALLATION STAGE, IT IS THE RESPONSIBILITY OF THE DESIGNER OR SYSTEM OWNER/OPERATOR TO SPECIFY THE LOCATIONS FOR, AND SIZES OF, ACCESS DOOR INSTALLATION TO ENSURE THAT, AT THE POINT OF HANDOVER, FULL SAFE ACCESS FOR CLEANING INSPECTION AND MAINTENANCE IS ACHIEVED IN LINE WITH THE REQUIREMENTS OF THE CONSTRUCTION DESIGN AND MANAGEMENT (CDM) REGULATIONS. FULL 'AS INSTALLED' DRAWINGS SHOULD BE PRODUCED AND BE AVAILABLE VIA THE CLIENT TO CLEANING/MAINTENANCE CONTRACTORS.
- FIRE DAMPERS MUST NOT BE FITTED IN NEW INSTALLATIONS. WHERE FIRE DAMPERS ARE STILL FOUND IN OLDER INSTALLATIONS, THESE SYSTEMS SHOULD BE TESTED ANNUALLY, AS A MINIMUM, AS REQUIRED BY BS 9999 - FIRE SAFETY IN THE DESIGN, MANAGEMENT AND USE OF BUILDINGS CODE OF PRACTICE, OR REMOVED FROM THE SYSTEM ENTIRELY. REMOVAL WILL REQUIRE A REVIEW OF THE FIRE RISK ASSESSMENT.
- OTHER SERVICES WHICH MAY BE INSTALLED AFTER THE ORIGINAL INSTALLATION OF THE EXTRACT DUCTWORK, SUCH AS ARCHITECTURAL FEATURES AND BUILDING FABRIC SHOULD NOT OBSTRUCT ANY ACCESS PANELS PROVIDED TO ALLOW FOR REGULAR CLEANING AND MAINTENANCE. EQUALLY ANY KITCHEN EQUIPMENT INSTALLED BELOW ANY DUCT ACCESS PANELS SHALL NOT RESTRICT ACCESS FOR CLEANING PURPOSES AND WHERE NECESSARY SHALL BE OF AN EASILY MOVABLE NATURE ON FLEXIBLE SERVICES CONNECTIONS.
- BESPOKE PERMANENTLY FIXED ACCESS GANTRIES OR PLATFORMS MAY BE REQUIRED ABOVE CEILINGS TO ENABLE SAFE ACCESS TO CLEANING ACCESS DOORS FOR HIGH-LEVEL WORKING. THESE MUST BE SPECIFIED BY THE DESIGNER UNDER THE CDM REGULATIONS IN CONSULTATION WITH THE OWNER/OPERATOR PRIOR TO SITE HANDOVER.

- PLEASE NOTE THAT THE INFORMATION ON THIS DOCUMENT IS NOT EXHAUSTIVE PROVIDING ONLY A SUMMARY OF THE MAIN ITEMS IN THE STANDARDS.
- ALL INSTALLING DUCTWORK CONTRACTORS SHALL ENSURE THAT THEIR INSTALLATION COMPLIES FULLY WITH THE REQUIREMENT OF DW144 - SHEET METAL DUCTWORK & DW172 KITCHEN VENTILATION SYSTEMS.
- WITH REGARD TO ACCESS HATCHES THE INSTALLATION SHALL FULLY COMPLY WITH THE REQUIREMENT OF DW144 - SHEET METAL DUCTWORK, TR19 INTERNAL CLEANLINESS OF VENTILATION SYSTEMS AND TR19 GREASE - FIRE RISK MANAGEMENT OF GREASE ACCUMULATION WITHIN KITCHEN EXTRACT SYSTEMS.
- WHEN INSTALLING PLASTIC DUCTWORK, THE INSTALLATION SHALL FULLY COMPLY WITH THE REQUIREMENTS OF DW154 - SPECIFICATION FOR PLASTIC DUCTWORK.

DW144 - DUCTWORK PRESSURE CLASSES

DUCT PRESSURE CLASS	STATIC PRESSURE LIMIT (PA)		MAXIMUM AIR VELOCITY(M/S)	AIR LEAKAGE LIMITS LITRES PER SECOND PER SQUARE METRE OF DUCT SURFACE AREA (DP=DIFFERENTIAL PRESSURE IN PASCALS)
	POSITIVE	NEGATIVE		
1	2	3	4	5
LOW PRESSURE - CLASS A	500	500	10	0.027 x dp ^{0.65}
MEDIUM PRESSURE - CLASS B	1000	750	20	0.009 x dp ^{0.65}
HIGH PRESSURE - CLASS C	2000	750	40	0.003 x dp ^{0.65}
HIGH PRESSURE - CLASS D	2000	750	40	0.001 x dp ^{0.65}

DW144 - DUCTWORK SUPPORT SPACING

TABLE 16	SUPPORTS FOR RECTANGULAR HORIZONTAL DUCTWORK				
LONGEST SIDE	HANGERS		BEARERS		MAXIMUM SPACING'S
	SCREWED ROD	FLAT STRAPS	STIRRUPS	"Z" cm" x-x	
UP TO 400	6	25 x 0.8 PLAIN OR PERFORATED	N/A	0.4	3000
401-600	8	30 x 3 FLAT	N/A	0.7	3000
601-1000	8	N/A	N/A	1.5	3000
1001-1500	10	N/A	N/A	2.1	2500
1501-2000	10	N/A	N/A	3.05	2500
2001-3000	12	N/A	N/A	4.45	2500

TABLE 17	SUPPORTS FOR CIRCULAR HORIZONTAL DUCTWORK				
DIAMETER	HANGERS		BEARERS		MAXIMUM SPACING'S
	SCREWED ROD (mm)	FLAT STRAPS	STIRRUPS	"Z" cm" x-x	
UP TO 315	6	25 x 0.8 PLAIN OR PERFORATED	25 x 3 FLAT	N/A	3000
316-500	8	N/A	30 x 3 FLAT	N/A	3000
501-800	8	N/A	30 x 3 FLAT	N/A	3000
801-1120	10	N/A	40 x5 FLAT	N/A	3000
1221-1500	10	N/A	40 x 5 FLAT	N/A	3000
1501-2400	12	N/A	50 x 6 FLAT	N/A	3000

TABLE 18	SUPPORTS FOR FLAT OVAL HORIZONTAL DUCTWORK				
MAJOR AXIS	HANGERS		BEARERS		MAXIMUM SPACING'S
	SCREWED ROD (mm)	FLAT STRAPS	STIRRUPS	"Z" cm" x-x	
UP TO 400	6	25 x 0.8 PLAIN OR PERFORATED	25 x 3 FLAT	0.4	3000
401-605	8	25 x 3 FLAT	30 x 3 FLAT	0.7	3000
606-1040	8	30 x 3 FLAT	30 x 3 FLAT	1.5	3000
1041-1510	10	N/A	N/A	2.1	3000
1511-1785	10	N/A	N/A	3.05	3000

Notes to Tables 16 to 18

- THE MAXIMUM SPACING'S SET OUT IN TABLES 16 TO 18 ARE RELATED SOLELY TO DUCT WEIGHT CONSIDERATIONS. CLOSER SPACING MAY BE REQUIRED BY REASON OF THE LIMITATIONS OF THE BUILDING STRUCTURE OR TO ACHIEVE THE NECESSARY DUCT RIGIDITY.
- AS AN ALTERNATIVE TO SCREWED ROD, WIRE ROPE MAY BE UTILISED TO SUIT INDIVIDUAL MANUFACTURER'S FIXING METHODS AND LOADING LIMITATIONS. SEE FIGS 167 TO 172.
- TABLE 19 OPPOSITE INDICATES TYPICAL ELASTIC SECTION MODULI FOR COMMONLY USED BEARER SECTION FOR RECTANGULAR AND FLAT OVAL DUCTWORK.

NOMINAL DUCTWORK WEIGHTS - CIRCULAR

Ductwork Weights - Circular

Ø mm	WEIGHT PER LENGTH (KILOGRAMS)		Total +	Insulation			Total
	1 METRE	GAUGE	kg/m	kg/m 3	Thickness	Weight	kg/m
2000	72.45		86.94	80	50	25.12	117.08
1600	57.97		69.56	80	50	20.10	93.68
1500	54.35		65.22	80	50	18.84	87.83
1400	50.73		60.88	80	50	17.58	81.98
1250	45.28		54.34	80	50	15.70	73.18
1120	40.58	1.2mm	48.70	80	50	14.07	65.58
1000	29.13		34.96	80	50	12.56	50.03
900	25.58		30.70	80	50	11.30	44.26
800	18.2		21.84	80	50	10.05	33.90
710	16.15		19.38	80	50	8.92	30.08
630	14.34		17.21	80	50	7.91	26.70
600	13.66		16.39	80	50	7.54	25.44
560	12.73		15.28	80	50	7.03	23.72
500	11.37		13.64	80	50	6.28	21.18
450	10.24		12.29	80	50	5.65	19.07
400	9.1		10.92	80	50	5.02	16.95
355	8.08	0.8mm	9.70	80	50	4.46	15.05
315	5.38		6.46	80	50	3.96	11.20
300	5.13		6.16	80	50	3.77	10.68
250	4.27		5.12	80	50	3.14	8.89
224	3.83		4.60	80	50	2.81	7.97
200	3.42	0.6mm	4.10	80	50	2.51	7.12
160	2.28		2.74	80	50	2.01	5.15
150	2.14		2.57	80	50	1.88	4.83
125	1.78	1	2.14	80	50	1.57	4.02
100	1.43	0.5mm	1.72	80	50	1.26	3.22

Information based on Lindab information

NOTES:
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2. DO NOT SCALE FROM THIS DRAWING. WORK TO FIGURED DIMENSIONS ONLY. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS STATED OTHERWISE.
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VENTILATION SERVICES NOTES

- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH OTHER PROJECT DOCUMENTATION INCLUDING EQUIPMENT SCHEDULES
- FLEXIBLE DUCTWORK SHALL BE USED FOR FINAL CONNECTIONS ONLY. NO FLEXIBLE DUCTWORK SHALL BE MORE THAN 500mm IN LENGTH.
- ALL DUCTWORK SHALL BE MANUFACTURED & INSTALLED IN ACCORDANCE WITH HVAC DW144, DW172 & DW154.
- ALL KITCHEN EXTRACT OR MOISTURE LADEN EXTRACT SYSTEM TO BE INSTALLED IN LINE WITH THE REQUIREMENTS OF DW172.
- DUCTWORK SHALL HAVE ACCESS PANELS IN ACCORDANCE WITH THE REQUIREMENTS OF DW144 AND TR19 AS REQUIRED..
- THERMAL INSULATION THICKNESS SHALL BE IN ACCORDANCE WITH RELEVANT BRITISH STANDARD BS5422
- ALL PLANT AND EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS INSTRUCTIONS
- VCD TO BE INSTALLED INLINE WITH BSRIA BG45/2015 INCLUDING ALL FINAL RUN OUTS
- FRESH AIR & EXHAUST TERMINALS SHALL BE A MINIMUM OF 2m APART. WHERE THE RISER TO ROOF IS LESS THAN THIS THE DUCTWORK SHALL ROUTE HORIZONTALLY AT ROOF LEVEL TO ACHIEVE THE MINIMUM DISTANCE.
- ALL PENETRATIONS TO BE PROVIDED BY THE MAIN CONTRACTOR/ BUILDER
- PLEASE REFER TO FOLLOWING DOCUMENTS FOR DETAILS OF FIRE DAMPERS.
 - KEY-XX-D0C-M-57-001 - FIRE DAMPER TYPICAL DETAIL - ACTIONAIR (SWECON)
 - KEY-XX-D0C-M-57-002 - FIRE & SMOKE DAMPER TYPICAL DETAIL - ACTIONAIR (SWECON)

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